

# Software Requirements Specification (SRS)

*Data Science Capstone Project*

## Handwritten Answer Evaluation System

### 1. Introduction

**Purpose:** This document describes the requirements for the Handwritten Answer Evaluation System.

**Scope:** The system converts scanned/photographed handwritten answer sheets into digital text using OCR, compares them against a model answer using NLP, and generates marks and feedback automatically — supporting questions of different mark values and optional/choice questions.

**Definitions:**

- SRS – Software Requirements Specification
- OCR – Optical Character Recognition
- NLP – Natural Language Processing
- Rubric – A marking scheme broken into weighted key points
- ASAG – Automated Short Answer Grading

### 2. Project Overview

**Project Title:** Handwritten Answer Evaluation Using OCR and NLP

**Domain:** Education Technology (EdTech) / Academic Assessment

**Description:**

This project reads handwritten answer sheets, digitizes them using OCR, and uses NLP and a trained machine learning model to compare each answer against a reference answer/rubric. It produces marks and written feedback per question, correctly handling variable marks per question and optional/choice question formats.

### 3. Tech Stack

- Language: Python 3.13
- Image Processing: OpenCV, Pillow, PyMuPDF
- OCR: EasyOCR, Tesseract (pytesseract)
- NLP: Sentence-Transformers, scikit-learn (TF-IDF), NLTK
- Machine Learning: scikit-learn (Random Forest), pandas, PyArrow, joblib
- Web Framework: Streamlit
- Dataset: ASAG2024 (Hugging Face)

### 4. Business Background

Manual evaluation of handwritten answer sheets is slow, inconsistent across evaluators, and does not scale well to large classes.

**Pain Points:**

- Time-consuming manual evaluation of large batches of answer sheets
- Inconsistent partial-credit scoring across evaluators
- Difficulty correctly handling optional/choice questions (e.g., attempt 4 of 5)
- No automatic check for poor-quality (blurry) scanned uploads
- Limited, delayed feedback provided to students

## 5. User Roles

### Teacher / Examiner:

- Upload the question paper and answer key
- Review generated marks and feedback

### Student:

- Submit a scanned/photographed handwritten answer sheet
- View marks and point-by-point feedback

### System Administrator:

- Maintain the system, OCR engines, and trained models
- Manage datasets used for model training

## 6. Functional Requirements

**FR-1 Answer Sheet Upload** — Upload a scanned image or multi-page PDF of a handwritten answer sheet.

**FR-2 Image Quality Check** — Detect blurry uploads and reject them, prompting re-upload.

**FR-3 OCR Extraction** — Convert handwritten answer text into digital text.

**FR-4 NLP Comparison** — Compare the extracted answer with the model answer using semantic similarity and keyword matching.

**FR-5 Rubric-Based Scoring** — Assign partial credit per key point, scaled to each question's marks.

**FR-6 Choice Question Handling** — Correctly grade optional/choice questions (attempt N of M, internal OR) without exceeding the paper's maximum marks.

**FR-7 Batch Grading** — Grade multiple students' answer sheets against the same question paper and answer key in one pass.

**FR-8 Feedback Generation** — Generate marks and written, point-by-point feedback for each answer.

## 7. Non-Functional Requirements

- Performance – Process and score an answer sheet within a reasonable time.
- Security – Protect uploaded student answer sheets and personal data.
- Scalability – Support grading an entire class/batch in one session.
- Reliability – Guarantee awarded marks never exceed a question's or paper's declared maximum.
- Usability – Simple upload-based interface requiring no coding knowledge from teachers or students.

## 8. Assumptions

- Answer sheets are legibly handwritten.
- The question paper and answer key are available in text form.
- Users have basic familiarity with uploading files via a web interface.

## 9. Constraints

- OCR accuracy depends on handwriting legibility and scan/photo quality.
- Limited project duration restricts training-data coverage across all subjects.
- Scoring model performance depends on the availability of labeled training data.

## 10. Use Case Summary

| Use Case            | Description                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------|
| Answer Sheet Upload | Upload a scanned/photographed handwritten answer sheet (image or multi-page PDF).            |
| Blur Check          | Validate image sharpness and reject blurry uploads before processing.                        |
| OCR Extraction      | Convert handwritten answer text into digital text.                                           |
| NLP Comparison      | Compare the extracted answer against the model answer using similarity and keyword matching. |
| Rubric Scoring      | Assign partial-credit marks per key point, per question.                                     |
| Batch Grading       | Grade an entire class's answer sheets against the same question paper and answer key.        |

## 11. Future Enhancements

- A handwriting-specific OCR model (e.g., TrOCR) for improved accuracy on cursive writing
- Support for additional subjects and regional languages
- A mobile app for direct answer-sheet capture
- Automated model retraining as new graded data becomes available
- A real-time grading dashboard for institutions

## Conclusion

The Handwritten Answer Evaluation System will help institutions automate the grading of handwritten answer sheets using OCR and NLP, improving evaluation speed and consistency while correctly supporting real exam formats such as variable marks per question and optional/choice questions.